Demo for Data and Network Security report

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Abstract—The abstract goes here.

Index Terms—Machine Learning, Deep Learning, Spying People?...

1 Introduction

In this section you need to introduce the topic, explain the problem you want to tackle and why it is something interesting to treat, the contribution you want to propose and why your proposal is better than current state-of-the-art. Is the project academic or industrially oriented?

2 Background

2.1 Deep Learning Networks

Complex computational problems can be addressed through various methodologies, prominent among which is Deep Learning. This paradigm is fundamentally rooted in mathematical optimization. Consider a predictive model defined as:

$$\begin{aligned} &f_\theta : \mathbb{R}^K \to \mathbb{R}^m \\ &\forall x \in \mathbb{R}^K, \Theta(x) \in \mathbb{R}^m \end{aligned} \tag{1}$$

The objective of training is to optimize the internal parameters—specifically the weights and biases of the neurons—to maximize the model's predictive accuracy.

This optimization process is systematically executed through the following iterative phases:

- Initialization: The network parameters are initially set using randomized values or predefined distribution strategies.
- Forward Propagation: An input vector x is processed through the mapping function to generate a predicted output, denoted as $\hat{y}$ (commonly referred to as the forward pass).
- Error Evaluation: The discrepancy between the predicted output and the ground-truth label y is quantified by evaluating a Loss function $\mathbf{L}$ (defined in Equation 2).
- Backpropagation and Optimization: Utilizing the backpropagation algorithm, the gradient of the loss function is computed with respect to each network weight via the chain rule. The model parameters are subsequently adjusted in the negative gradient direction—a process known as gradient descent—effectively minimizing the objective function.

$$\begin{aligned} \hat{y} &= \Theta(x) \\ L(x) &= y - \hat{y} \end{aligned} \tag{2}$$

While the mathematical formulation of the loss function varies depending on the specific task architecture, its underlying objective remains constant: the resulting scalar serves as a performance metric utilized by optimization algorithms to converge toward the global minimum[1].

The foundational principles detailed above describe standard Machine Learning. The distinction between classical Machine Learning and Deep Learning primarily lies in structural complexity; standard Machine Learning models typically rely on shallow architectures, whereas Deep Learning utilizes deep neural networks characterized by multiple successive hidden layers.

Formally, $\text{Deep Learning} \subset \text{Machine Learning}$. While both paradigms maintain distinct domains of utility, this report focuses primarily on the deployment and behavior of Deep Learning architectures.

2.2 Base Station

This infrastructure comprises cellular base stations maintained by telecommunication service providers. These stations aggregate vast quantities of telemetry and location data which, despite the application of anonymization techniques, remain vulnerable to re-identification attacks orchestrated via deep learning models. Consequently, any adversary possessing sufficient computational resources could exploit these vulnerabilities to execute large-scale mass surveillance[2].

3 Overview of your proposed approach

In this section you need to provide scientific and technical detail about your proposal, explaining what you want to do, describing the idea and how it differs to current state-of-the-art. How can your project be done? Select approaches to form the basis of the solution, ensure you can justify it

4 Evaluation

In this section you should propose an experiment to evaluate the goodness of your proposal. Of course for this report you are NOT REQUIRED TO DO ANY EXPERIMENTS, it's just a PROPOSAL of experiments that may be useful to do.

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1. The parameter configuration that minimizes the loss value.
2. Amnesty International: An Easy Guide to Mass Surveillance

5 Related work

In this section, you need to analyze your direct competitors (e.g. [1] if you talk about watermarking), explaining a bit what they do and why you think your approach is better. What has been tried before?

6 Conclusions

This section concludes the report providing a brief summary of what can be accomplished within the project and possible further future extensions.

PS: It is mandatory to include the section with bibliographic references in this research report

References

[1] Y. Adi, C. Baum, M. Cisse, B. Pinkas, and J. Keshet, "Turning your weakness into a strength: Watermarking deep neural networks by backdooring," in 27th USENIX Security Symposium (USENIX Security 18). Baltimore, MD: USENIX Association, Aug. 2018, pp. 1615–1631.